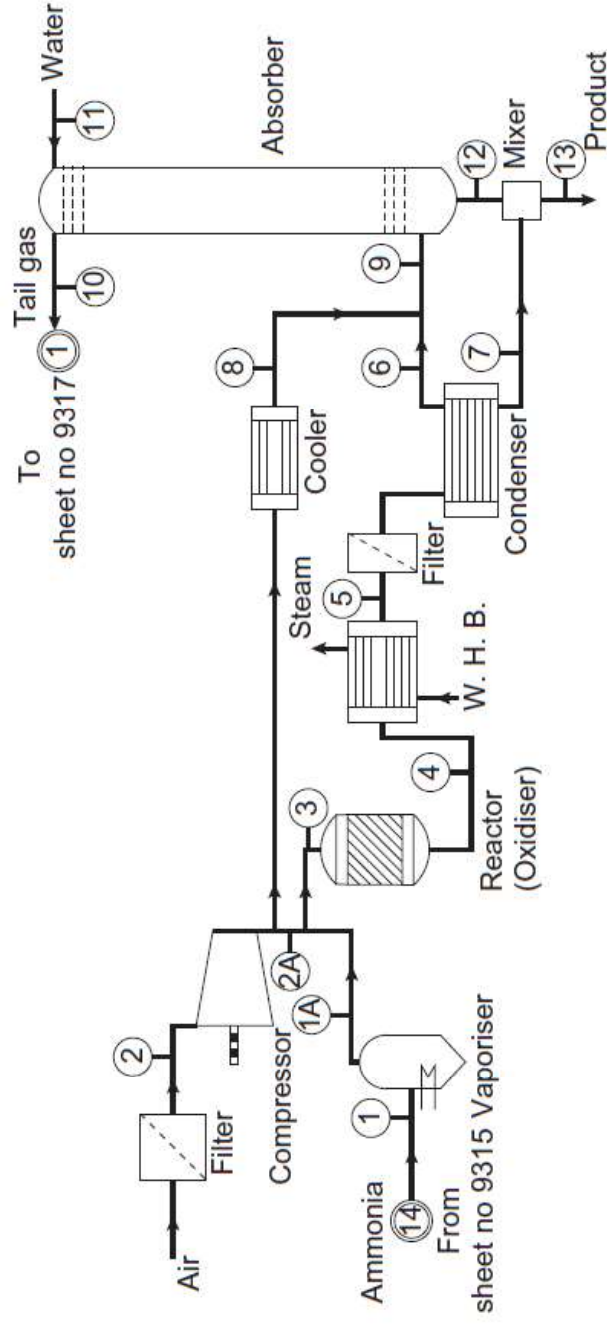




Flows kg/h pressures nominal

Line no. Stream component	1 Ammonia feed	1A Ammonia vapor	2 Filtered air	2A Oxidiser air	3 Oxidiser feed	4 Oxidiser outlet	5 W.H.B. outlet	6 Condenser gas	7 Condenser acid	8 Secondary air	9 Absorber feed	10 Tail(2) gas	11 Water feed	12 Absorber acid	13 Product acid	C & R Construction Inc
NH ₃	731.0	731.0	—	—	731.0	Nil	—	—	—	—	—	—	—	—	—	Nitric acid 60 percent
O ₂	—	—	3036.9	2628.2	2628.2	935.7 (935.7) ⁽¹⁾	275.2	Trace	Trace	408.7	883.9	371.5	—	Trace	Trace	100,000 ty
N ₂	—	—	9990.8	8644.7	8644.7	8668.8	8668.8	Trace	Trace	1346.1	10,014.7	10,014.7	—	Trace	Trace	Client BOP chemicals
NO	—	—	—	—	—	1238.4 (1238.4) ⁽¹⁾	202.5	—	—	—	202.5	21.9	—	Trace	Trace	SLIGO
NO ₂	—	—	—	—	—	Trace (7) ⁽¹⁾	967.2	—	—	—	967.2	(Trace) ⁽¹⁾	—	Trace	Trace	Sheet no. 9316
HNO ₃	—	—	—	—	—	Nil	—	850.6	—	—	—	—	—	Trace	Trace	
H ₂ O	—	—	Trace	—	—	1161.0	29.4	1010.1	—	—	29.4	26.3	1376.9	1704.0	2554.6	
Total	731.0	731.0	13,027.7	11,272.9	12,003.9	12,003.9	10,143.1	1860.7	1754.8	11,897.7	10,434.4	1376.9	2840.0	4700.6	—	
Press bar	8	8	1	8	8	8	8	1	8	8	8	1	8	1	1	Dwg by
Temp. °C	15	20	15	230	204	907	40	40	40	40	40	25	25	40	43	Date
																Checked
																25/7/1980

FIGURE 2.8

Flowsheet: simplified nitric acid process.